A · PATIENT DEMOGRAPHICS & ELBOW PRESENTATION PROFILE — INTAKE

Patient Name _____ Age _____ Sex ☐ M ☐ F Date _____

Occupation / Throwing or Manual Sport _____ Dominant Arm ☐ R ☐ L Affected Side ☐ R ☐ L ☐ Bilat Onset ☐ Traumatic ☐ Overuse / Insidious

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Compartment ☐ Lat ☐ Med ☐ Ant ☐ Post Worst _____

B · POST-TRAUMA RADIOGRAPHY RULES, DISTAL BICEPS AVULSION & RED-FLAG TRIAGE — TRIAGE GATE

ELBOW EXTENSION TEST & TRAUMA ROM RULES — Appelboam 2008

- ☐ **Elbow extension test** (Appelboam 2008 BMJ, adults + children, n = 1740) [H]: **fracture: Sn 96.8% (95.0–98.2), Sp 48.5% (45.6–51.4); worst-case Sn 95.3%; Sp similar in adults (47.7%) and children (49.5%)**
- ☐ **Elbow extension test — adults** (Appelboam 2008 BMJ) [H]: **full extension: NPV 98.4% (96.3–99.5), LR− 0.03 (0.01–0.08); radiography can be deferred if an olecranon fracture is not suspected; patients not X-rayed "should return if symptoms have not resolved within 7-10 days"**
- ☐ **Range of movement and the radiograph** (Lennon 2007 Emerg Med J) [M]: **for detecting a normal radiograph: full extension Sp 0.916 (0.863–0.969), Sn 0.478; equal ROM Sp 0.976 (0.940–0.991), Sn 0.211; under 16 years equal ROM Sp 1 (0.941–1.000)**
- ☐ **Active ROM after trauma** (Darracq 2008 Am J Emerg Med, n = 113) [M]: **limited active ROM (flexion, extension, pronation, supination): Sn 100% (0.93–1.00) for fracture or effusion; preserved active ROM: Sp 97% (0.89–1.00) for absence of fracture**

DISTAL BICEPS AVULSION & EMERGENCY RED FLAGS — O'Driscoll 2007

- ☐ **Hook test — distal biceps avulsion** (O'Driscoll 2007 Am J Sports Med, n = 45, surgery reference) [M]: **Sn and Sp both 100%; MRI Sn 92%, Sp 85%** · **Biceps squeeze test** (Ruland 2005 Clin Orthop Relat Res) [L]: **positive in 24 of 25 patients; complete rupture confirmed at surgery in 21 of 22**
- ☐ **Dislocation & compartment syndrome: dislocation with neurovascular compromise** (absent pulse, pallor, acute nerve deficit); **forearm compartment syndrome** (severe pain out of proportion, tense swollen forearm, pain on passive finger stretch) [E]
- ☐ **Septic & paediatric red flags: septic olecranon bursitis / septic arthritis** (fever, heat, swelling); **fracture in a child** (acute trauma, refusal to use arm, swelling or incomplete active ROM) [E]
- ☐ **Visceral & proximal referral: cardiac referral to the left medial arm** (non-mechanical ache, chest/jaw symptoms) [E] · **cervical radiculopathy referral** (C5–T1 neck/radicular screen) [E]

TRIAGE DISPOSITION: ☐ **Cleared — proceed** ☐ Radiography required (incomplete extension / trauma / child) ☐ Urgent surgical referral (distal biceps avulsion) ☐ **Emergency referral (dislocation / compartment syndrome / septic / cardiac)**

C · CLINICAL PATHWAY ROUTER & ELBOW SUBGROUP NAVIGATION — RAPID

CLINICAL PATHWAY ROUTER — Tick route → jump to matching pages

☐ A · TENDON: 1 → 2 → 3 → 4 → 8 ☐ B · INSTABILITY: 1 → 2 → 4 → 8 → 9 ☐ C · POST-OP: 1 → 3 → 5 → 8 → 9

D · SUBJECTIVE HISTORY, SYMPTOM BEHAVIOUR & ELBOW / FOREARM BODY CHART — SUBJECTIVE

SYMPTOM LOCATION & CLINICAL PATTERN — Clinical Signpost

- ☐ **Lateral Epicondyle Pain on Gripping / Wrist Extension:** (+) Cozen / PFG → **Lateral epicondylalgia / LET (p2/p4/p8)**
- ☐ **Medial Epicondyle Pain or Valgus Throwing Overload:** Medial test / Moving valgus → **Medial epicondylalgia vs UCL tear (p2/p4)**
- ☐ **Medial Ache with Ring / Little Finger Paraesthesia:** (+) Flexion + pressure / Tinel → **Cubital tunnel syndrome (p2/p4/p7)**
- ☐ **Lateral Clicking / Apprehension in Supinated Push-Up:** (+) Chair / push-up / tabletop → **Posterolateral rotatory instability (p2/p4)**

AGGRAVATING, EASING & IRRITABILITY PROFILE — Behaviour

Aggravating _____ Easing _____

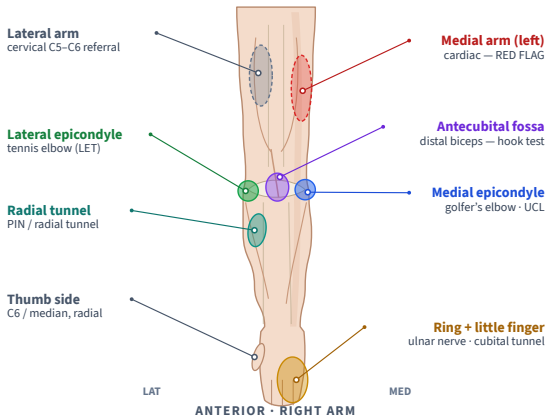
Irritability: ☐ Low ☐ Moderate ☐ High (rest/night pain)

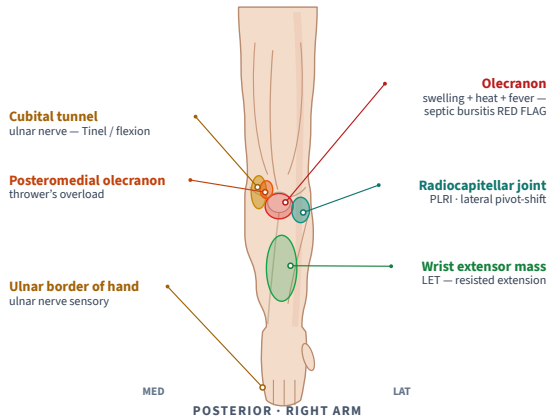
MECHANISM & OCCUPATIONAL / SPORT RISK — History

Mechanism: ☐ FOOSH / trauma ☐ Eccentric load (biceps) ☐ Overhead throwing ☐ Repetitive + forceful arm use

Factors: ☐ Smoking history ☐ Prior CSI injection ☐ Locking / catching ☐ Ulnar / radial paraesthesia

BODY CHART — WHERE IS THE PAIN? — Clinical signposts [E]





Mark on the chart: shade = pain · //// = pins & needles · ×× = numbness · → = radiation · write 0–10 beside each area

Unable to fully extend after trauma → elbow extension test (X-ray). **Pop/ecchymosis in antecubital fossa** → hook test.

SUBJECTIVE HISTORY · MECHANISM · PATIENT GOALS (PSFS)

A · ELBOW DIFFERENTIAL DIAGNOSIS MATRIX (LATERAL, MEDIAL, ANTERIOR & POSTERIOR COMPARTMENTS)

DIFFERENTIAL

CONDITION	HALLMARK HISTORY & PAIN LOCATION	KEY PHYSICAL FINDINGS & PROVOCATION	EVIDENCE & ATLAS
Lateral epicondylalgia (LET)	Lateral epicondyle pain aggravated by gripping & repetitive + forceful arm activity (OR 5.6, 1.9–16.5)	Tender common extensor origin; pain on Cozen ▸, Mill ▸, Maudsley ▸, chair test; reduced pain-free grip	H Prevalence 1.3% (Shiri 2006)
Medial epicondylalgia	Insidious medial epicondyle ache with forceful wrist flexion, pronation, or gripping	Tender common flexor-pronator origin; pain on medial epicondylitis test ▸; stable to valgus stress	H Prevalence 0.4% (Shiri 2006) · E
Cubital tunnel syndrome ▸	Medial elbow ache with paraesthesia/numbness in ring & little fingers, worse with sustained elbow flexion	(+) Pressure + flexion ▸ (Sn 0.91 at 30 s), Tinel ▸ (Sn 0.70 / 54%, NPV 98%), scratch collapse ▸ (Sn 69%), ULTT3 ▸	M Novak 1994 · L Cheng 2008
Radial tunnel / PIN	Deep aching pain over mobile wad / supinator mass distal to lateral epicondyle (or pure motor finger-extensor weakness in PIN)	Maximal tenderness over radial tunnel distal to lateral epicondyle; pain with resisted supination / Maudsley ▸ & ULTT2b ▸	E Clinical differentiation
UCL injury ▸	Overhead thrower or acute valgus trauma; medial pain during late cocking / acceleration phase or pop	(+) Moving valgus stress test ▸ (Sn 100% / 98%, pain arc 120° to 70°, max 90°) & milking manoeuvre ▸ (LR+ 5.1, Sn 77%)	M Sciascia 2026 · L O'Driscoll 2005
Posterolateral rotatory instability (PLRI) ▸	Lateral clicking, locking, or slipping after prior elbow dislocation, trauma, or lateral surgery	(+) Chair / push-up / tabletop ▸ (chair and push-up signs each 7 of 8 positive), lateral pivot-shift ▸ (3/8 awake), posterolateral drawer ▸	L Regan 2006
Distal biceps rupture ▸	Sudden eccentric extension load against flexed elbow; audible pop, antecubital ecchymosis, proximal belly retraction	Abnormal hook test ▸ (Sn 100%, Sp 100%); (+) biceps squeeze test ▸ (24/25); weakness in supination > flexion	M O'Driscoll 2007 · L Ruland 2005
OCD capitellum	Adolescent thrower or gymnast with lateral elbow pain, effusion, catching, or locking on valgus loading	Radiocapitellar joint-line tenderness, loss of terminal extension, crepitus or mechanical catching	E Clinical differentiation
OA / stiffness	Post-traumatic contracture or degenerative joint pain at terminal flexion and extension	Hard bony or capsular end-feel limiting AROM ▸ / PROM ▸ flexion, extension, pronation, or supination	E Clinical differentiation

B · PREVALENCE, RISK FACTORS & NATURAL COURSE VS CORTICOSTEROID INJECTION / PHYSIOTHERAPY

PROGNOSIS

EPIDEMIOLOGY & RISK FACTORS (SHIRI 2006)

Shiri 2006

- ☐ **Prevalence (age 30–64, n = 4,783)** (Shiri 2006 Am J Epidemiol) H: **definite lateral epicondylitis 1.3%, medial 0.4%; highest at 45–54 years; no sex difference**
- ☐ **Lateral epicondylitis — risk factors** (Shiri 2006 Am J Epidemiol) H: **current smoking OR 3.4 (1.4–8.3); former smoking OR 3.0 (1.3–6.6); repetitive + forceful arm activity OR 5.6 (1.9–16.5)**

NATURAL COURSE VS INJECTION & PHYSIOTHERAPY RCTS

Bisset 2006

- ☐ **Natural course vs treatment (RCT, n = 185)** (Smidt 2002 Lancet) H: **success at 6 weeks: injection 92%, physiotherapy 47%, wait-and-see 32%; at 52 weeks: injection 69%, physiotherapy 91%, wait-and-see 83%**
- ☐ **MWM + exercise vs injection vs wait-and-see (RCT, n = 198)** (Bisset 2006 BMJ) H: **at 52 weeks injection worse than physiotherapy (0.3, 0.1 to 0.5; NNT 4) and than wait-and-see on 2 of 3 measures (0.3, 0.04 to 0.4; NNT 4); after injection 47/65 (72%) deteriorated after 3 or 6 weeks**

C · DIAGNOSTIC ACCURACY SUMMARY: CUBITAL TUNNEL, ULNAR COLLATERAL LIGAMENT & PLRI

ACCURACY

CUBITAL TUNNEL PROVOCATIVE TESTS (NOVAK 1994 · CHENG 2008)

Ulnar Nerve

- ☐ **Cubital tunnel provocative tests ▸** (Novak 1994 J Hand Surg Am, 32 patients, EMG-proven) M: **Tinel Sn 0.70; at 30 seconds: elbow flexion 0.32, pressure 0.55, pressure + flexion 0.91 (most sensitive)**
- ☐ **Scratch collapse test ▸** (Cheng 2008 J Hand Surg Am) L: **cubital tunnel: Sn 69% (Tinel 54%, flexion/compression 46%), accuracy 89%; Tinel NPV 98%; carpal tunnel Sn 64%**

PLRI APPREHENSION SIGNS (REGAN 2006)

LUCL / PLRI

- ☐ **PLRI apprehension signs (n = 8) ▸** (Regan 2006 J Shoulder Elbow Surg) L: **chair sign positive in 7, push-up sign positive in 7; pivot-shift positive awake in 3 (all under anaesthesia); chair + push-up more sensitive than awake pivot-shift**

UCL MOVING VALGUS & MILKING TESTS (O'DRISCOLL 2005 · SCIASCIA 2026)

Medial / UCL

- ☐ **UCL — moving valgus stress test ▸** (O'Driscoll 2005 Am J Sports Med, n = 21, surgery / arthroscopy reference) L: **Sn 100% (17/17), Sp 75% (3/4); pain arc 120° to 70°, maximum at 90°**
- ☐ **UCL — moving valgus vs milking ▸** (Sciascia 2026 Am J Sports Med) M: **moving valgus stress test Sn 98%, accuracy 93% (best to rule out); milking manoeuvre LR+ 5.1, Sn 77% (best to rule in); combining tests added nothing**

Primary Working Diagnosis: _____

Secondary / Contributing Factors: _____

DIFFERENTIAL REASONING · PROGNOSTIC COUNSELLING · RISK-FACTOR MODIFICATION NOTES

Clinician _____ **Date** _____

Underline = Atlas card · ▸ = video · Evidence: H high M moderate L low E expert · Confidential · v2026.10

A · ELBOW & FOREARM RANGE OF MOTION WRITE-IN TABLE & GONIOMETRIC REAL-CHANGE THRESHOLDS

GONIOMETRY

MOTION (AROM ▸ / PROM ▸)	AFFECTED AROM	AFFECTED PROM	UNAFFECTED	END-FEEL / PAIN	ARMSTRONG 1998 REAL CHANGE [M]
Elbow Flexion				☐ Soft ☐ Caps ☐ Pain	Same tester **> 6°** · Between testers **10°**
Elbow Extension				☐ Bony ☐ Caps ☐ Pain	Same tester **> 7°** · Between testers **10°**
Forearm Pronation				☐ Firm ☐ Pain	Same tester **> 8°** · Between testers **10°**
Forearm Supination				☐ Firm ☐ Pain	Same tester **> 8°** · Between testers **11°**
Wrist Flex / Ext (CPG)				☐ Full ☐ Pain	Baseline & follow-up / discharge (Lucado 2022)

☐ **Goniometry (universal goniometer, same tester)** (Armstrong 1998 J Shoulder Elbow Surg) [M]: **real change if greater than 6° flexion, 7° extension, 8° pronation, 8° supination; between testers 10° flexion, 10° extension, 10° pronation, 11° supination**

B · PAIN-FREE GRIP, MAXIMUM GRIP & JOSPT LET CPG PHYSICAL IMPAIRMENT MEASURES

DYNAMOMETRY

GRIP STRENGTH DYNAMOMETRY & CHANGE SCORES

Grip Calc

☐ **Grip strength in LET (interobserver)** (Smidt 2002 Arch Phys Med Rehabil) [M]: **ICC pain-free grip .97, maximum grip .98, severity of complaints .90, pressure pain threshold .77 (unsatisfactory)**

☐ **Pain-free grip — change scores** (Sveinall 2025 Shoulder Elbow) [M]: **SDC95 6.4 kg pain-free grip, 8.4 kg maximum grip; MIC 6.5 kg pain-free, 1 kg maximum; pain on gripping SDC95 4, MIC 3.5 (0–10)**

JOSPT LET CPG CORE MEASURES (LUCADO 2022)

JOSPT 2022

☐ **JOSPT LET CPG — measures** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should include the physical impairment measures of elbow and wrist range of motion, pressure pain threshold, pain-free grip strength, and maximum grip strength at baseline and at least one other follow-up point that includes discharge"**

MEASURE	AFFECTED	UNAFFECTED	RATIO (%)	VERIFIED CHANGE BENCHMARK (SVEINALL 2025)
Pain-Free Grip (kg)				**SDC95 6.4 kg · MIC 6.5 kg** (ICC .97, Smidt 2002)
Maximum Grip (kg)				**SDC95 8.4 kg · MIC 1 kg** (ICC .98, Smidt 2002)
Pain on Gripping (0–10) & PPT				**Pain on gripping SDC95 4 · MIC 3.5 (0–10)** · PPT ICC .77

C · REGIONAL PALPATION MAP, CARRYING ANGLE, C5–T1 NEUROLOGICAL SCREEN & ULTT2B / ULTT3

NEURO & PALP

COMPARTMENT PALPATION MAP & ALIGNMENT [E]

Lateral: ☐ Lat epicondyle / ECRB ☐ Radial head ☐ Radial tunnel / supinator ☐ Soft-spot effusion
Medial: ☐ Med epicondyle / flexor-pronator ☐ Sublime tubercle / UCL
☐ Cubital tunnel / ulnar n. subluxation
Ant / Post: ☐ Distal biceps tendon ☐ Brachialis ☐ Olecranon tip / bursa ☐ Triceps insertion
Carrying angle: ☐ Symmetric ☐ Cubitus valgus ☐ Cubitus varus **Notes:** _____

RADIAL & ULNAR NEURODYNAMIC TESTING [E]

☐ **ULTT2b (radial nerve / PIN bias)** ▸ [E]: shoulder depression, elbow extension, shoulder IR + forearm pronation, wrist/finger flexion; structural differentiation via cervical lateral flexion

☐ **ULTT3 (ulnar nerve / cubital tunnel bias)** ▸ [E]: shoulder depression, wrist/finger extension, forearm pronation, elbow flexion, shoulder ER + abduction; structural differentiation

ULTT2b L / R: _____ **ULTT3 L / R:** _____

ROOT [E]	MYOTOME KEY ACTION	DERMATOME SENSORY ZONE	DEEP TENDON REFLEX	FINDING (L / R)
C5	Shoulder abduction / elbow flexion	Lateral arm (regimental badge)	Biceps	
C6	Elbow flexion / wrist extension	Lateral forearm, thumb & index finger	Brachioradialis	
C7	Elbow extension / wrist flexion	Middle finger & dorsal hand	Triceps	
C8	Thumb extension / finger flexion	Medial forearm, ring & little fingers	—	
T1	Finger abduction / adduction (interossei)	Medial elbow & proximal forearm	—	

OBJECTIVE EXAMINATION · GONIOMETRY · GRIP DYNAMOMETRY · NEURODYNAMIC NOTES

A · EPICONDYLALGIA, COLLATERAL LIGAMENT (UCL / LCL) & POSTEROLATERAL ROTATORY INSTABILITY SPECIAL TESTS

SPECIAL TESTS

TEST & CATEGORY	TECHNIQUE & POSITIVE CRITERION	VERIFIED DIAGNOSTIC PERFORMANCE	RESULT
Cozen's test ▸ (LET)	Pronated fist, radial deviation + resisted wrist extension → lateral epicondyle pain	Provocative contractile test [E]	☐ – ☐ +
Mill's test ▸ (LET)	Passive forearm pronation, full wrist/finger flexion + passive elbow extension → lateral pain	Passive extensor stretch test [E]	☐ – ☐ +
Maudsley's ▸ **& chair test** (LET)	Resisted middle-finger extension (Maudsley) or pronated 3-finger chair lift → lateral epicondyle pain	Extensor digitorum / ECRB provocation [E]	☐ – ☐ +
Medial epicondylitis test ▸	Palpate medial epicondyle; passive forearm supination + elbow/wrist extension or resisted flexion/pronation	Flexor-pronator origin provocation [E]	☐ – ☐ +
Valgus ▸ **& varus stress tests** ▸	Static valgus (UCL) or varus (LCL) torque with elbow unlocked from extension → joint gapping or pain	Static collateral laxity assessment [E]	☐ – ☐ +
Moving valgus stress test ▸ (UCL)	Abducted/ER shoulder; constant valgus torque while extending from full flexion; medial pain in **120° to 70°** arc (max **90°**)	**Sn 100% (17/17), Sp 75% (3/4)** (O'Driscoll 2005) [L] · **Sn 98%, accuracy 93% (best to rule out)** (Sciascia 2026) [M]	☐ – ☐ +
Modified milking manoeuvre ▸ (UCL)	Pull thumb laterally with forearm supinated & elbow flexed to apply valgus load across UCL	**LR+ 5.1, Sn 77% (best to rule in)**; combining tests added nothing (Sciascia 2026) [M]	☐ – ☐ +
Lateral pivot-shift ▸ **& drawer** ▸ (PLRI)	Supine overhead arm: axial compression + valgus + supination (pivot-shift) or posterolateral translation (drawer)	Pivot-shift positive awake in **3 of 8** (all under anaesthesia) (Regan 2006) [L] · Drawer [E]	☐ – ☐ +
Chair, push-up & tabletop relocation ▸ (PLRI)	Supinated chair rise, push-up, or tabletop press-up relieved by radial head thumb stabilisation	**Chair sign positive in 7 of 8, push-up sign positive in 7 of 8**; more sensitive than awake pivot-shift (Regan 2006) [L]	☐ – ☐ +

B · CUBITAL TUNNEL & PERIPHERAL NERVE PROVOCATION TESTS (NOVAK 1994 · CHENG 2008)

NERVE TESTS

NERVE TEST	PROTOCOL & POSITIVE FINDING	VERIFIED ACCURACY (NOVAK 1994 · CHENG 2008)	RESULT
Tinel sign at elbow ▸	Percussion over ulnar nerve in retrocondylar groove / cubital tunnel → ring/little finger tingling	**Sn 0.70** (Novak 1994) [M] · **Sn 54%, NPV 98%** (Cheng 2008) [L]	☐ – ☐ +
Elbow flexion & pressure + flexion ▸	Sustained maximal elbow flexion, direct ulnar nerve pressure, or combined pressure + flexion for **30 seconds**	At 30 s: **elbow flexion Sn 0.32, pressure 0.55, pressure + flexion 0.91 (most sensitive)** (Novak 1994) [M] · Flexion/compression **46%** (Cheng 2008) [L]	☐ – ☐ +
Scratch collapse test ▸	Cutaneous scratch over cubital tunnel followed immediately by resisted bilateral shoulder ER → unilateral give-way	Cubital tunnel: **Sn 69%, accuracy 89%**; carpal tunnel **Sn 64%** (Cheng 2008) [L]	☐ – ☐ +

C · VALIDATED ELBOW PROM DASHBOARD (PRTEE, OXFORD ELBOW SCORE, QUICKDASH, KJOC, MEPS, NPRS, PSFS)

PROMS

OUTCOME SCALE	SCORING RULE & RANGE	VERIFIED RELIABILITY, SDC / MDC & MIC / MCID	INITIAL	FOLLOW-UP
PRTEE Calc	15 items (0–10): Pain (5 items, 0–50) + Function (10 items ÷ 2, 0–50) = Total 0–100	Test-retest **0.96 (0.94–0.97)**, SEM **3.1 (1.8 to 4.4)**, MDC95 **8.9 (5.3–12.5)** (Shafiee 2022) [M] · Tennis elbow: total **SDC95 17, MIC 9**; pain **SDC95 8, MIC 11**; function **SDC95 12, MIC 4** (Sveinall 2025) [M]		
Oxford Elbow Score (OES) Calc	12 items (0–4), 3 domains (Function, Pain, Social-Psych, 4 items each; sum ÷ 16 × 100 = 0–100)	After surgery: **MCID ≈ 10 for Function (similar to DASH), ≈ 18 for Pain and Social-Psychological** (Dawson 2008) [M] · Elbow injury: **SDC 12.1** (Wänström 2024) [M]		
QuickDASH Calc	11 items (1–5, ≥ 10 answered): ((sum ÷ n) − 1) × 25 (0–100)	Elbow injury: **SDC QuickDASH 11.4, SANE 1.94** (Wänström 2024) [M]		
KJOC Score	Overhead throwers (0–100, higher = better performance)	Correlates with DASH (**−.693**) and DASH sports (**−0.804**); **threshold 81.3 separates category 1 from 3** (Domb 2010) [L]		
MEPS · NPRS · PSFS Calc	MEPS (pain, arc, stability, ADL) · NPRS (0–10) · PSFS (mean 0–10)	Clinical tracking scales [E]; pain on gripping (0–10) **SDC95 4, MIC 3.5** (Sveinall 2025) [M]		

SPECIAL TESTS SYNTHESIS · BASELINE PROM SCORES · FUNCTIONAL GOAL SETTING

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

A · POST-SURGICAL REHABILITATION: DISTAL BICEPS REPAIR, UCL RECONSTRUCTION / REPAIR & LET SHAM TRIAL

POST-OP

POST-OP RESTRICTIONS ARE SURGEON-DIRECTED [E]

DISTAL BICEPS REPAIR (AMARASOORIYA 2020 · AMIN 2016 · BERGMAN 2021)

Distal Biceps

- ☐ **Complications (SR)** (Amarasooriya 2020 Am J Sports Med) [M]: **major complications 4.6%; minor 20.4% (n = 630); most common lateral (antebrachial) cutaneous nerve injury 9.2% (n = 283); LABCN injury higher with limited anterior vs extensile single incision (9.7% vs 5.2%, P = .03); superficial radial nerve injury higher with extensile vs limited incision (6% vs 2.1%, P = .002); fixation technique had no effect on rerupture or PIN injury**
- ☐ **Single vs double incision (MA)** (Amin 2016 Orthop J Sports Med) [M]: **LABCN neurapraxia most common after single incision: 77 of 785 (9.8%)**
- ☐ **Early motion vs immobilisation (RCT, cortical button)** (Bergman 2021 J Bone Joint Surg Am) [H]: **no difference in return to work (p ≥ 0.83); "No clinically important group differences were seen"**

SURGERY VS SHAM FOR LET (KROSLAK 2018)

LET Surgery

- ☐ **Surgery vs sham for LET (RCT, n = 26)** (Kroslak 2018 Am J Sports Med) [H]: **no significant difference in any parameter at any stage; > 6500 patients per group would be needed to show a difference**

UCL RECONSTRUCTION & INTERNAL BRACE REPAIR

Erickson 2015

- ☐ **UCL reconstruction (SR, 20 studies, 2019 athletes)** (Erickson 2015 Orthop J Sports Med) [M]: **pooled return to sport 86.2%; 90% excellent/good (Conway-Jobe); RTS collegiate 95.5% vs high school 89.4% vs professional 86.4%; docking 97.0%, ASMI 93.3%, Jobe 66.7%**
- ☐ **UCL reconstruction — large series** (Cain 2010 Am J Sports Med, 743 athletes ≥ 2 y) [L]: **83% returned to the same or higher level in less than 1 year**
- ☐ **UCL repair with internal brace** (Dugas 2019 Am J Sports Med, 111 throwers) [L]: **92% (102/111) returned to the same or higher level at a mean 6.7 months; mean KJOC 88.2**
- ☐ **UCL reconstruction — rehab & RTS criteria (SR, 33 studies, 3480 athletes)** (Kemler 2022 Am J Sports Med) [M]: **RTS timelines 6.5 to 16 months; protocols most often used bracing with progressive ROM (93.9%), strengthening (84.8%) and an interval throwing programme (81.8%); no study fully defined RTS criteria**

B · POST-TRAUMATIC STIFFNESS / ARTHROLYSIS, HETEROTOPIC OSSIFICATION & RADIAL HEAD FRACTURES

TRAUMA & STIFFNESS

POST-TRAUMATIC STIFFNESS: ARTHROLYSIS & SPLINTING

Lanzerath 2023

- ☐ **Arthrolysis (SR, 30 studies, 798 pts)** (Kodde 2013 J Shoulder Elbow Surg) [M]: **ROM gain 51°, 40°, 88°, 56° and complications 23, 5, 73, 58% for open, arthroscopic, open + external fixator, open + distraction arthroplasty**
- ☐ **Arthrolysis (SR, 27 studies, 1666 pts)** (Lanzerath 2023 Arch Orthop Trauma Surg) [M]: **success (MEPS excellent/good): open 88.8% (revision 6.3%, other complications 18.1%); arthroscopic 91.8% (revision 1.6%, complications 9.1%)**
- ☐ **Splinting for post-traumatic stiffness (RCT, n = 66)** (Lindenhovius 2012 J Bone Joint Surg Am) [H]: **no difference between static progressive and dynamic splinting; stiffness "can improve with exercises and dynamic or static splinting over a period of six to twelve months, and patience is warranted"**

HETEROTOPIC OSSIFICATION (SHUKLA 2015 · KHAN 2019)

HO Risk

- ☐ **Heterotopic ossification** (Shukla 2015 J Shoulder Elbow Surg) [L]: **after surgical fracture-dislocation 12 of 28 elbows (43%); 7 of 9 (77%) after multiple closed reductions**
- ☐ **HO after total elbow arthroplasty (SR)** (Khan 2019 J Shoulder Elbow Surg) [M]: **radiographic 10%, symptomatic 3%**

RADIAL HEAD & MASON I–II FRACTURE TRIALS

Harding 2011

- ☐ **Radial head fracture — early mobilisation (RCT, n = 60)** (Liow 2002 Injury) [M]: **immediate mobilisation: less pain and better function at 1 week, no harm**
- ☐ **Radial head fracture — protocols (RCT, n = 180)** (Paschos 2013 J Orthop Trauma) [M]: **early mobilisation safe and effective; a delay of 48 hours before early mobilisation "could be advantageous"**
- ☐ **Elbow fractures — early mobilisation (Cochrane)** (Harding 2011 Cochrane) [L]: **no difference between early and delayed mobilisation in pain or ROM limitation (Mason I–II); all participants returned to full ADL, none changed occupation**
- ☐ **Mason II — operative vs non-operative (RCT)** (Mulders 2021 J Shoulder Elbow Surg) [M]: **similar function at 1 year**

C · WOUND / NEURO / HO RE-CHECK BOX & OPERATIVE DETAILS LOG

OPERATIVE LOG

Post-Op Re-Check [E]: ☐ Wound erythema / drainage / infection ☐ LABCN / superficial radial / PIN / ulnar deficit ☐ Suspected heterotopic ossification (loss of arc + warmth) ☐ Graft / button / hardware failure

Procedure / Graft / Fixation _____ **Surgery Date** _____ **Surgeon** _____

Surgeon-Directed Restrictions & Splint / Hinge Settings _____

Ulnar Nerve / LABCN / PIN Status _____ **Target RTS / RTW Milestone** _____

POST-OPERATIVE COMMUNICATION · SURGEON ORDERS · SPLINTING PROGRESSION NOTES

A · CORTICOSTEROID, PRP & AUTOLOGOUS BLOOD INJECTIONS IN LATERAL EPICONDYLALGIA

INJECTIONS

CORTICOSTEROID INJECTION (BISSET 2006 · COOMBES 2010 & 2013)

Coombes 2013

- ☐ **Corticosteroid injection (Bisset)** (Bisset 2006 BMJ) [H]: **better at 6 weeks, but "paradoxically reversed after six weeks, with high recurrence rates"; worse than physiotherapy at 52 weeks (NNT 4) — see page 2**
- ☐ **Corticosteroid injection (SR, 41 RCTs)** (Coombes 2010 Lancet) [H]: **lateral epicondylalgia: large short-term pain effect vs no intervention (SMD 1·44, 1·17–1·71) but no intervention favoured at intermediate term (−0·40, −0·67 to −0·14) and long term (−0·31, −0·61 to −0·01); serious adverse event (tendon rupture) in 1 of 991 (0·1%)**
- ☐ **Corticosteroid vs placebo injection ± physio (RCT, n = 165)** (Coombes 2013 JAMA) [H]: **1 year: recovery 83% vs 96% (RR 0.86, 99% CI 0.75–0.99); recurrence 54% vs 12% (RR 0.23, 0.10–0.51); 26 weeks: 55% vs 85% (RR 0.79, 0.62–0.99)**

PRP, AUTOLOGOUS BLOOD & NON-OPERATIVE MA

Karjalainen 2021

- ☐ **PRP / autologous blood (Cochrane, 8 studies)** (Karjalainen 2021 Cochrane) [H]: **vs placebo at 3 months: pain 0.16 points better (0–10), function 2 points (0–100) — not clinically important**
- ☐ **Non-operative treatments (MA)** (Kim 2021 JSES Int) [M]: **PRP vs control: function SMD 0.14 (−0.45 to 0.73); autologous blood vs control: function −0.07 (−0.64 to 0.50); strengthening vs no treatment: pain MD −0.07 (−0.56 to 0.41), function SMD −0.08 (−0.46 to 0.30); "does not support the use of nonoperative treatment options including physiotherapy (strengthening), CSI, PRP, or AB"**

B · ESWT, LASER, ORTHOSES, TAPING & JOSPT 2022 LET CPG MODALITIES RECOMMENDATIONS

EPAS & ORTHOSES

ESWT, LOW-LEVEL LASER & ORTHOSES REVIEWS

Bjordal 2008

- ☐ **ESWT (Cochrane)** (Buchbinder 2005 Cochrane) [M]: **one trial (93 participants): steroid injection may be more effective than ESWT**
- ☐ **Low-level laser (MA)** (Bjordal 2008 BMC Musculoskelet Disord) [M]: **vs control: pain WMD 10.2 mm (3.0 to 17.5) on 100 mm VAS; 904 nm directly on the tendon (5 trials): 17.2 mm (8.5 to 25.9) more than placebo; pain-free grip SMD 0.66 (0.42 to 0.90)**
- ☐ **Orthoses / counterforce brace (Cochrane)** (Struijs 2002 Cochrane) [L]: **"No definitive conclusions can be drawn concerning effectiveness of orthotic devices"**

JOSPT LET CPG — ORTHOSES & TAPING (LUCADO 2022)

JOSPT 2022

- ☐ **Counterforce / wrist orthosis** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians may use a forearm counterforce or wrist support orthosis to be worn during activity for immediate improvement of pain and strength in those with LET whose symptoms are aggravated with activity"**; for intermediate or long-term symptoms: **"a recommendation cannot be made"**
- ☐ **Rigid & kinesiology taping** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should use rigid taping techniques for immediate/short-term pain relief and improvement in pain-free muscle function in those with irritable LET"**; **"Clinicians may use kinesiology tape application as part of a multimodal treatment program for immediate and short-term management of pain and muscle function"**

JOSPT LET CPG — ELECTROPHYSICAL AGENTS (LUCADO 2022)

JOSPT 2022

- ☐ **Ultrasound** (Lucado 2022 JOSPT LET CPG) (Guideline): **"a recommendation cannot be made for the use of ultrasound as a stand-alone treatment"**
- ☐ **Phonophoresis** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should not use phonophoresis with 10% hydrocortisone gel, topical prednisolone (2 mg/d), or 1% diclofenac sodium gel"**
- ☐ **Iontophoresis** (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use iontophoresis with an anti-inflammatory drug, early in the rehabilitation phase (no later than 2–4 weeks from onset or aggravation of symptoms), in individuals presenting with highly irritable symptoms"**
- ☐ **Laser therapy** (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use laser therapy for improvements in pain and grip strength, seen in follow-up periods >4 weeks to 6 months"**
- ☐ **Cryotherapy + burst TENS** (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use cryotherapy combined with burst transcutaneous electrical nerve stimulation (TENS) to reduce pain in the short term in individuals with symptoms of LET for greater than 30 days"**

C · ORTHOSIS, TAPING & ELECTROPHYSICAL AGENT DELIVERY LOG

PARAMETER LOG

DATE	MODALITY / ORTHOSIS / TAPE	SITE, WAVELENGTH / DOSE / TENSION	PRE / POST PFG OR NPRS	PATIENT RESPONSE

INJECTION COUNSELLING · TAPING / ORTHOSIS FIT · MODALITY NOTES

Clinician _____ Date _____

A · MOBILISATION WITH MOVEMENT (MWM), JOINT MOBILISATION & DEEP TRANSVERSE FRICTION EVIDENCE

MANUAL THERAPY

MOBILISATION WITH MOVEMENT (VICENZINO 2001 · COOMBES 2013)

Coombes 2013

- ☐ **MWM (lateral glide) — crossover RCT** (Vicenzino 2001 Man Ther) [M]: **specific, immediate hypoalgesia on the affected side**
- ☐ **MWM + exercise (Coombes 2013)** (Coombes 2013 JAMA) [H]: **at 4 weeks, placebo injection + physiotherapy: recovered/much improved 39% vs 10% without physio (RR 4.00, 1.07–15.00); at 1 year physio vs no physio 91% vs 88% (RR 1.04, 0.90–1.19), recurrence 29% vs 38%**

JOINT MOBILISATION SR & DEEP TRANSVERSE FRICTION

Lucado 2019

- ☐ **Joint mobilisation (SR / MA)** (Lucado 2019 J Hand Ther) [M]: **"compelling evidence that joint mobilizations have a positive effect on both pain and/or functional grip scores across all time frames compared to control groups"**
- ☐ **Deep transverse friction (Cochrane)** (Loew 2014 Cochrane) [L]: **uncertain whether it improves pain and function (very low-quality evidence)**

B · JOSPT LET CPG MANUAL THERAPY & DRY NEEDLING QUOTES + CUBITAL TUNNEL CONSERVATIVE CARE

CPG & NEURAL

JOSPT LET CPG — MANUAL THERAPY & DRY NEEDLING (LUCADO 2022)

JOSPT 2022

- ☐ **Local elbow manipulation / mobilisation** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should use local elbow joint manipulation or mobilization techniques to reduce pain and increase pain-free grip strength … for those who can tolerate the specific technique"**
- ☐ **Cervical, thoracic & wrist adjuncts** (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use manipulation or mobilization techniques directed at the cervical spine, thoracic spine, and/or wrist as an adjunct to local treatment for short-term pain relief … when impairments in those regions are identified"**
- ☐ **IASTM & deep transverse friction** (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use instrument-assisted soft tissue mobilization combined with exercise to improve pain and function in those with chronic LET"**; deep transverse friction: **"a recommendation cannot be made"**
- ☐ **Dry needling** (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should use either tendon or trigger point dry needling for the treatment of pain and functional deficits associated with LET"**

CUBITAL TUNNEL SYNDROME CONSERVATIVE CARE

[E]

- ☐ **Patient education & posture modification** [E]: avoid sustained elbow flexion, leaning on the medial epicondyle / retrocondylar groove, and repetitive flexion-extension under load
- ☐ **Night splinting / extension towel wrap** [E]: maintain comfortable elbow extension during sleep to reduce nocturnal ulnar nerve traction and intraneural pressure
- ☐ **Neural mobilisation** [E]: gentle, non-provocative ulnar nerve gliding exercises; monitor intrinsic hand strength and refer if progressive motor deficit develops

C · PRE-INTERVENTION SAFETY SCREEN, INFORMED CONSENT & MANUAL THERAPY / DRY NEEDLING LOG

CONSENT

Contraindications / Precautions Screened [E]: ☐ Fracture / unreduced dislocation excluded ☐ No acute HO / myositis ossificans ☐ No septic bursitis / overlying infection ☐ Anticoagulation / needle precautions checked

CONSENT OBTAINED FOR: ☐ Elbow MWM / joint mobilisation ☐ Cervical / thoracic / wrist adjunct ☐ Tendon / trigger-point dry needling ☐ IASTM / neural mobilisation

Patient / Guardian Signature _____ **Clinician** _____ **Date** _____

DATE	TECHNIQUE (MWM / JOINT MOB / NEEDLING / NEURAL)	SEGMENT / DIRECTION / DOSAGE	IMMEDIATE PFG / ROM	ADVERSE EVENTS

MANUAL THERAPY RESPONSE · MWM GLIDE VECTOR · HOME SELF-MWM INSTRUCTIONS

Clinician _____ **Date** _____

A · JOSPT LET CPG EXERCISE RECOMMENDATIONS, STRENGTHENING MA, UCL & STIFFNESS REHABILITATION

EXERCISE & RTS

JOSPT LET CPG — EXERCISE (LUCADO 2022)

JOSPT 2022

- ☐ **Wrist extensor loading** ▸ (Lucado 2022 JOSPT LET CPG) (Guideline): **"Clinicians should use isometric, concentric, and/or eccentric therapeutic resisted exercises of the wrist extensors in the treatment of individuals with subacute or chronic LET"**
- ☐ **Multimodal combination** ▸ (Lucado 2022 JOSPT LET CPG) (Guideline): **"should use therapeutic resisted wrist extension strengthening exercises in combination with other therapeutic interventions, including manual therapy"**
- ☐ **Phased return & kinetic chain** ▸ (Lucado 2022 JOSPT LET CPG) (Guideline): **"may use a phased approach to reintroduce stress, increase strength, improve endurance, and restore optimal motor control"** in high-demand occupations, hobbies, performing arts or sport; **"may include shoulder and scapular stabilizer muscle training exercises, when impairments are identified"**

STRENGTHENING MA, NATURAL COURSE, UCL & STIFFNESS

Evidence

- ☐ **Strengthening vs no treatment (MA)** ▸ (Kim 2021 JSES Int) [M]: **no difference in pain (MD −0.07) or function (SMD −0.08) — see page 6**
- ☐ **Natural course comparison** ▸ (Smidt 2002 Lancet · Coombes 2013 JAMA) [H]: **at 52 weeks physiotherapy 91% vs wait-and-see 83% success (Smidt 2002, page 2); MWM + exercise vs no physio 91% vs 88% at 1 year (Coombes 2013)**
- ☐ **UCL non-operative rehab** (Erickson 2015 Sports Health) [L]: **"Nonoperative treatment should be attempted, but limited studies have not shown promising results"** · **UCL post-op** (Kemler 2022) [M]: **interval throwing programme in 81.8% of protocols; RTS 6.5 to 16 months**
- ☐ **Post-traumatic stiffness** (Lindenhovius 2012) [H]: **exercise + splinting improves over six to twelve months** · **Medial epicondylalgia, loading specifics & RTW:** graded flexor-pronator loading, isometric/eccentric/heavy slow resistance progression & ergonomic task modification [E]

B · 12-WEEK LATERAL / MEDIAL ELBOW PROGRESSIVE LOADING LOG

12-WEEK LOG

EXERCISE / DRILL	WK 1	WK 2	WK 3	WK 4	WK 5	WK 6	WK 7	WK 8	WK 9	WK 10	WK 11	WK 12
Isometric Wrist Ext / Flex												
Concentric / Eccentric Wrist Ext												
Supination / Pronation Hammer Load												
Grip / Carry Progression												
Scapular & Rotator Cuff Kinetic Chain												

C · INTERVAL THROWING PROGRAMME LOG (FLAT-GROUND & MOUND PROGRESSION)

THROWING LOG

DATE / PHASE	DISTANCE / SURFACE	THROWS & EFFORT	WARM-UP / CROW-HOP	MEDIAL PAIN / NEXT-DAY SORENESS	PROGRESSION DECISION

HOME EXERCISE PRESCRIPTION · OCCUPATIONAL / SPORT CONDITIONING · INTERVAL THROWING NOTES

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____

#	DATE	NPRS	FLEX / EXT	PRON / SUP	PFG / MAX (KG)	EXERCISE & LOADING DELIVERED	MANUAL / TAPE / EPA	IRRITABILITY	PROM	INIT
1						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
2						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
3						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
4						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
5						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
6						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
7						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
8						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
9						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
10						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
11						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		
12						Isom | Conc/Ecc | Sup/Pron | Scap | Throw	MWM | Joint Mob | Needling | Tape	Low | Mod | High		

DISCHARGE MILESTONES (POPULATION-SPECIFIC MIC / MCID & SDC / MDC BENCHMARKS) & DISCHARGE SUMMARY

DISCHARGE

- ☐ **PRTEE** [M]: **Tennis elbow** (Sveinall 2025): total (0–100) **SDC95 17, MIC 9**; pain (0–50) **SDC95 8, MIC 11**; function (0–50) **SDC95 12, MIC 4** · **SR** (Shafiee 2022): **MDC95 8.9 (5.3–12.5), SEM 3.1 (1.8 to 4.4)**
- ☐ **Oxford Elbow Score** [M]: **After surgery** (Dawson 2008): **MCID ≈ 10 for Function, ≈ 18 for Pain and Social-Psychological** · **Elbow injury** (Wänström 2024): **SDC 12.1**
- ☐ **QuickDASH & SANE** [M]: **Elbow injury** (Wänström 2024): **SDC QuickDASH 11.4, SANE 1.94**
- ☐ **Pain-free & maximum grip** [M]: **Tennis elbow** (Sveinall 2025): **SDC95 6.4 kg pain-free grip, 8.4 kg maximum grip; MIC 6.5 kg pain-free, 1 kg maximum; pain on gripping SDC95 4, MIC 3.5 (0–10)**
- ☐ **Goniometry real change** [M] (Armstrong 1998): same tester **> 6° flex, 7° ext, 8° pron, 8° sup**; between testers **10° flex, 10° ext, 10° pron, 11° sup**
- ☐ **KJOC** [L]: **Throwers after UCL reconstruction** (Domb 2010): **threshold 81.3 separates category 1 from 3**

Final PRTEE / OES / QuickDASH _____ **Final PFG / Max Grip (kg)** _____ **Final Flex / Ext / Pron / Sup** _____ **Disposition** ☐ Self-Mgmt / RTS ☐ Ortho Refer

DISCHARGE SUMMARY · LONG-TERM RELAPSE PREVENTION · RETURN-TO-SPORT / WORK CLEARANCE

Discharging Clinician _____ **Sig.** _____